

Wi-Fi Security Protocol

Module 31

Duration: 25:36

Video: https://www.youtube.com/watch?v=MI3_13Hn9-I

Course Coordinator - Dr Maninder Singh, Professor and Dean, Academic Affairs, Thapar University, Patiala

Subject Expert - Prof Yogesh Chaba, Department of Computer Engineering and IT, Guru Jambheshwar University of Science & Technology, Hisar

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0:03 [Music]

0:24 [Music]

0:31 Hello friends I am Professor yogesh Chaba from Department of computer science and engineering Guru jambheshwar

0:40 University of Science and Technology Hisar harana today in this lecture we

0:46 will study about Wi-Fi technology first of all please see the

0:53 contents which I shall cover after covering brief introduction about Wi-Fi

1:00 technology I shall take up its layered protocol architecture which will be

1:06 followed by Wi-Fi standards in last I shall cover WiFi frame

1:14 format a wireless local area network W Lan is a data Communication System

1:22 implemented as an extension or as an alternative to a wired LAN WiFi is a

1:31 technology of Wireless local area network based on the i e

1:38 802.11 series of Wireless standards issued by the Institute of electrical and

1:45 electronic Engineers i e an international Professional Organization

1:53 for electrical and electronic Engineers the i e does not test equipment for

2:01 2 minutes, 1 second compliance with their standards the nonprofit Wi-Fi Alliance

2:09 was formed in 1999 to establish and enforce standards for interoperability and backward

2:18 compatibility and to promote Wireless local area Network Technology the Wi-Fi

2:27 Alliance enforces the use of the Wi-Fi brand to Technologies based on the i e

2:35 802.11 standard manufacturers with membership in the Wi-Fi Alliance whose

2:43 products are manufactured as per I E 802.11 standard gain the right to mark

2:52 their products with the Wii logo specifically the certification

2:59 process requires compliance to the i e 802.11 radio standards the

3:08 WPA and WPA2 security standards and the

3:14 EAP authentication standards any manufacturer who manufactures product as

per the standard can put a Wi-Fi logo on their product

there are several Technologies to choose from while selecting a wireless LAN solution each with advantages and limitations most Wireless lens use spread Spectrum a whiteband radio frequency technique in which the signal is spread out over the available bandwidth there are two types of spread Spectrum first is frequency hoping and second direct sequence frequency hoping spread

Spectrum FHS uses a narrow band carrier that changes frequency in a pattern known only to the transmitter and receiver on the other hand direct sequence spread Spectrum DS generates a red redundant bit pattern for each bit to be transmitted and requires more bandwidth for implementation this bit pattern called a chip or chipping code is used by the receiver to recover the original signal even if one or more bits in the chip are damaged during transmission statistical technique embedded in the radio can recover the original data without the need for retransmission now let us move on to the layered protocol architecture of i e 802.11 the i e 802.11 standard only covers the physical layer and medium excess layer of OSI reference model as shown in figure the physical layer is subdivided into physical layer convergence protocol plcp and the physical medium dependent sublayer PMD the plcp sublayer provides a carrier sense signal called Clear Channel assessment CCA and provides a common physical service access point sap independent of the transmission technology the second sublayer PMD handles modulations and encoding or decoding of signals the Mac management support medium Access Control Association and re Association of a station to an access point and roaming between different access points it also controls authentication mechanism encryption synchronization of a station with regard to an access point and power management to save battery power further 802.11 uses frequency hoping spread Spectrum fhws direct sequence spread Spectrum DS orthogonal frequency Division multiplexing ofdm and high rate direct sequence spread Spectrum HR DS modulation techniques at physical layer frequency hoping spread Spectrum FHS uses 79 channels of 1 mahz each and a pseudo random number generator is used to produce the sequence of frequencies hoped two this method provides a fair way to allocate Spectrum in the

7:52 unregulated ISM band direct sequence spread Spectrum technique support
8:01 8 minutes, 1 second communication speed of 1 or 2 Mbps it basically uses a
technique
8:09 similar to code division multiple access CDMA next modulation technique used in
8:17 physical layer is orthogonal frequency division multiplexing OFDM which
8:25 delivers 54 Mbps and above speed in The
8:30 Wider 5 GHz ISM band next technique is
8:36 OFDM that is high rate Orthogonal Frequency Sequence Spread Spectrum which offers speed of 11
8:47 and 22 Mbps in 2.4 GHz
8:54 bandwidth in MAC layer i.e. 802.11 uses two modes of operations the first
9:03 called distributed coordination function DCF and second Point coordination
9:10 function PCF DCF does not use any kind of central control it uses a protocol
9:19 called carrier sense multiple access CSMA with collision avoidance CSMA/CA in this
9:28 method both physical Channel sensing and virtual Channel sensing is done in
9:37 physical Channel sensing when a station wants to transmit it senses the channel
9:43 if it is Idle it just starts transmitting it does not sense the channel while
transmitting but transmits
9:53 full frame which may get destroyed at the receiver due to
10:00 interference if a collision occurs the colliding station waits for a random
10:06 amount of time using the ethernet binary exponential backoff algorithm and then
10:14 try again later in Virtual Channel sensing scheme medium access with collision
avoidance
10:23 CSMA/CA is used in point coordination scheme PCF base station is used to
10:31 control all activity in its cell now let us see different Wireless LAN standard
10:39 for i.e. 802.11 i.e. 802.11 is a set of medium access control
10:49 and physical layer specification for implementing Wireless local area network
10:57 in 2.4 and 5 GHz frequency
11:04 band they are created and maintained by the IEEE standard committee i.e.
11:15 802.11 the base version of the standard was released in
11:21 1997 and had many subsequent amendments the standard and amendments
11:29 provide the basis for wireless network products using the wifi brand there are
11:37 several Wireless LAN standards each suited for a particular environment as shown
in
11:46 figure first standard of IEEE 802.11 Series is
11:53 802.11 itself which was released in 1997
12:00 it offers 1 to 2 Mbps speed with an indoor range of 20 m and an outdoor
12:08 range of 100 m it uses

12:13 fhws and DS as spread Spectrum technique
 12:20 it operates on 2.4 GHz ISM band next standard is 82.
 12:29 11a which was released in 1999 it offers 54 and 8 Mbps speed with
 12:39 an indoor range of 35 M and outdoor range of 120 M it uses ofdm as
 12:48 modulation technique it operates on 5 GHz ISM band
 12:56 since the 2.4 GHz band is heavily used so using the relatively unused 5 GHz
 13:04 band gives 802.11a significant advantage in theory
 13:13 802.11a signals are absorbed more readily by walls and other solid objects
 13:22 in their path due to their smaller wavelength and as a result they cannot p
 13:29 penetrate now moving on to i.e. 802.11b this standard was released in
 13:39 1999 it offers 11 and 22 Mbps speed with
 13:45 an indoor range of 35 M and outdoor range of 120 M it uses HR
 13:55 DS as spread Spectrum technique it operates on
 14:01 14 minutes, 1 second 2.4 GHz ISM band next is i.e.
 14:08 802.11g standard this standard was released in 2003 it offers 54 Mbps speed
 14:18 with an indoor range of 38 M and outdoor range of 140 M it uses ofdm and DM as
 14:30 spread Spectrum technique it operates on 2.4 GHz ISM band i.e.
 14:40 802.11n released in 2009 is an amendment that improves upon the previous
 14:48 802.11 standards by adding multiple input multiple output mimo antennas
 14:57 802.11 and offer speed of 54 Mbps to 600 Mbps using ofdm
 15:07 spread Spectrum technique it operates on both the 2.4 GHz and the Lesser used 5
 15:16 GHz band support for 5 GHz band is optional this standard offers indoor
 15:25 range of 70 M and outdoor range of 250 m i.e.
 15:32 802.11 AC released in 2013 is built on
 15:40 802.11n it offers speed in the range of 1300 Mbps and operates on 5 GHz ISM
 15:52 band other i.e. 802.11 standards i.e. 802.11
 15:59 D addresses the spread of technology to countries not addressed by i.e. i.e.
 16:10 802.11e focuses on the quality of service associated with wireless
 16:16 transmission of multimedia applications i.e.
 16:23 802.11f addresses roaming between APS and interoperability between different
 16:32 vendor groups i.e. 802.11h focuses on frequency selection
 16:41 and power concerns on the 5 GHz band in European
 16:48 countries i.e. 802.11i focuses on enhancing WPA
 16:56 security and Authentication for
 17:00 802.11 that include radius karos and i.e.
 17:07 802.11x i.e. 802.11j focuses on the Japanese
 17:16 equivalent of 802.11h i.e.

17:22 802.11 a d is an amendment that defines a new physical layer for
17:30 802.11 Network to operate in the 60 GHz Spectrum products implementing the
17:40 802.11 ad standard are being brought to Market under the Y gig brand
17:49 name the certification program is now being developed by the wifi Alliance
17:56 instead of Y gig Alliance the peak transmission rate of
18:03 802.11 ad is 7 GBS per second i e
18:11 802.11 AF also referred to as whitei or super WiFi is an amendment
18:21 approved in February 2014 that allows W Lan operation in TV
18:28 white space Spectrum in the VHF and UHF
18:33 bands between 54 and 790
18:40 mahz it uses cognitive radio technology to transmit on unused TV channels
18:49 Wireless lens use electromagnetic waves to communicate information from one
18:56 point to another without relying on a wired connection radio waves are often
19:04 referred to as radio carriers because they simply perform the function of
19:10 delivering energy to a remote receiver the data being transmitted are
19:18 superimposed on the radio carrier so that they can be extracted accurately at
19:25 the receiving end this process is generally referred to as carrier
19:33 modulation i e 802.11b and
19:39 802.11g standard utilizes the 2.4 GHz Spectrum however
19:47 802.11a used the more heavily regulated 5 GHz band i e
19:56 802.11n utilizes both 2.4 GHz and 5 GHz
20:03 band these bands are commonly referred to as 2.4 GHz and 5 GHz band in most of
20:14 the literature each spectrum is subdivided into Channels with a center frequency
20:22 and bandwidth analog to the way radio and TV broadcast bands are
20:30 subdivided the 2.4 GHz band is divided into 14 channels spaced 5 mahz apart
20:39 beginning with Channel 1 which is centered on
20:45 2412 GHz the L channels have additional restrictions or are unavailable for use
20:54 in some regulatory domains figure shows exact frequencies
21:00 of 14 different channels used in 2.4 GHz ISM
21:09 band now we will move on to the frame format of I E
21:15 802.11 standard frame format of this Wireless lens standard is as shown in
21:23 figure there are nine fields in frame format of this standard starting from
21:31 frame control to checkum frame control field has 11
21:37 subfields now let us see each field one by one frame control field which is of
21:45 two bytes is used for control purpose details of subfield of frame
21:51 control field will be discussed after discussing main fields
21:59 second field is duration which tells how long the frame and its acknowledgement

22:06 will occupy the channel the frame header contains four
22:13 addresses the source and the destination are obviously needed and the other two
22:19 are used for the source and destination base stations for intercell traffic
22:30 the sequence field allows fragments to be numbered of the 16 bits available 12
22:38 identify the frame and four identify the fragment the data field contains the
22:46 payload up to 2312 bytes followed by the usual check sum of 4
22:54 bytes now let us see details of frame control field which has 11
23:02 subfields the first of these is protocol version which allows two versions of the
23:09 protocols to operate at the same time in the same cell then comes the type field
which
23:17 indicates data control or management type of frame and next field indicates
23:26 subtype the two DS and from DS bits indicates that the frame is going to or
23:35 coming from the inter cell distribution system for example ethernet
23:42 system the MF bit means that more fragments will follow the retry bit marks a retr
23:52 transmission of a frame sent earlier the power management
23:59 pwr bit is used by the base station to put the receiver into sleep state or
24:08 take it out of sleep State the more bit indicates that the
24:15 sender has additional frames for the receiver the wbit specifies that the
24:24 frame body has been encrypted using the web wired equivalent privacy algorithm
24:33 and finally the obit tells the receiver that a sequence of frame with this bit
24:41 on must be processed strictly in order so dear friends today in this
24:50 lecture we have covered introductory concepts of Wi-Fi technology followed by
24:56 its layered protocol architecture and i e
25:03 802.11 standards in last we have seen frame format of i e
25:11 802.11 technology so this was all about WiFi technology thank you
25:21 [Music]